DAY 1: STATIC ROUTING CONCEPTS

Theory + Cisco Packet Tracer Practical | Ahmedabad–Surat Branch Network

Theory

What is Static Routing? Static routing is a routing method in which a network administrator manually configures the path to a destination network. The router does not automatically learn or change the route using a dynamic routing protocol.

Advantages and Disadvantages

Type	Point	Real-world analogy
Advantage	Simple and predictable for small, stable networks.	A small Swiggy branch with fixed locations can use a known path.
Advantage	Low routing-protocol overhead because routers do not exchange route updates.	A fixed branch path avoids continuous routing-update traffic.
Disadvantage	Does not automatically adapt when a configured path fails.	If the Ahmedabad–Surat path fails, a manually configured route needs intervention or a configured backup.
Disadvantage	Manual configuration becomes difficult as the network grows.	An IRCTC-scale network with many locations would require many routes to be maintained manually.

Practical: Two-Router Static Routing

Goal: Configure two routers connected by a serial link so that an Ahmedabad LAN and a Surat LAN can communicate using static routes.

1. Topology and IP Addressing

PC1 — Switch1 — R1 (Ahmedabad) ===== Serial ===== R2 (Surat) — Switch2 — PC2

Device	Interface	IP Address	Subnet Mask
R1	G0/0	192.168.1.1	255.255.255.0
R1	S0/0/0	10.0.0.1	255.255.255.252
R2	S0/0/0	10.0.0.2	255.255.255.252
R2	G0/0	192.168.2.1	255.255.255.0
PC1	NIC	192.168.1.10	255.255.255.0
PC2	NIC	192.168.2.10	255.255.255.0

PC1 Default Gateway: 192.168.1.1 | PC2 Default Gateway: 192.168.2.1

2. Router1 (Ahmedabad) - Complete Commands

```
enable
configure terminal
hostname Ahmedabad

interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

interface serial 0/0/0
ip address 10.0.0.1 255.255.255.252
clock rate 64000
```

```
no shutdown
exit

ip route 192.168.2.0 255.255.255.0 10.0.0.2

end
write memory
```

3. Router2 (Surat) - Complete Commands

```
enable
configure terminal
hostname Surat

interface gigabitEthernet 0/0
ip address 192.168.2.1 255.255.255.0
no shutdown
exit

interface serial 0/0/0
ip address 10.0.0.2 255.255.255.252
no shutdown
exit

ip route 192.168.1.0 255.255.255.0 10.0.0.1

end
write memory
```

Note: Configure 'clock rate 64000' only on the DCE side of the serial connection. Your Packet Tracer router may use different interface names such as Fa0/0 or S0/3/0; verify with 'show ip interface brief'.

4. PC Configuration

PC	IP Address	Subnet Mask	Default Gateway
PC1 - Ahmedabad	192.168.1.10	255.255.255.0	192.168.1.1
PC2 - Surat	192.168.2.10	255.255.255.0	192.168.2.1

5. Testing and Verification

From Router1, test the PC connected to Router2:

```
Ahmedabad# ping 192.168.2.10

Ahmedabad# show ip route
Ahmedabad# show ip interface brief
```

From Ahmedabad PC1:

```
ping 192.168.2.10
```

From Surat PC2:

```
ping 192.168.1.10
```

If the configuration is correct and all interfaces are up, the PCs in both city LANs should communicate successfully.

Task-wise Answers

Task 1: Two routers are connected through a serial cable. R1 uses 10.0.0.1/30 and R2 uses 10.0.0.2/30 on the serial link. Each router also has a separate /24 LAN.

Task 2: Router1 uses the static route 'ip route 192.168.2.0 255.255.255.0 10.0.0.2'. Ping 192.168.2.10 from Router1 to verify reachability.

Task 3: R1 represents Ahmedabad and R2 represents Surat. Static routes are configured in both directions so 192.168.1.0/24 and 192.168.2.0/24 can communicate.

Task 4: Static routing is simple, predictable, and low-overhead for small networks, but it does not automatically adapt to failures and becomes difficult to maintain in large networks.